

Fourier

Fourier series

The Fourier series represents a periodic function as a sum of sine and cosine functions (or complex exponentials) with discrete frequencies.

Used to determine the frequency response of periodic signals.

Fourier transform

A generalization of the Fourier series.

Used to determine the frequency response of arbitrary complex signals.

Discrete Fourier Transform (DFT)

A Fourier transform for a discrete (sampled) signal.

Sample error add extra frequency components

Used for analyzing the frequency content of discrete signals.

Baseband and sideband

Baseband is the range of frequencies occupied by a signal that has NOT been modulated to higher frequencies. i.e. the highest frequency of original data.

Sideband is a band of frequencies at either side of the carrier frequency, that are the result of the modulation process.

Nyquist sampling limit: $f_s > 2 * f_n$

To sample baseband: double the frequency of baseband.

To sample sideband: cover both sideband frequency.

##

Continuous time is “**real time**”, where a signal can be defined **at arbitrary time**.

Discrete time is “**sampled time**”, where signal are measured a **specific** times and the information between samples can not be determined.